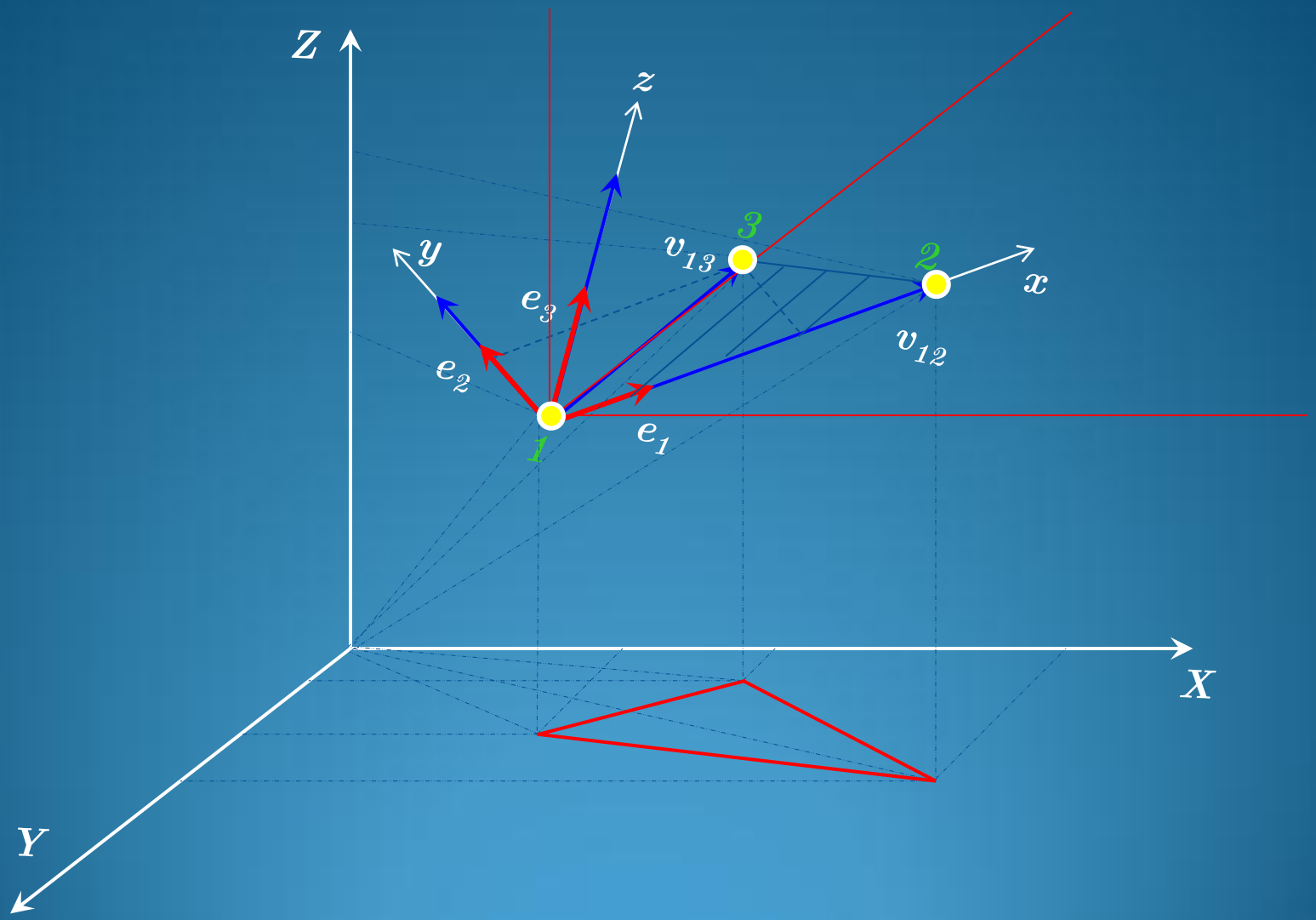
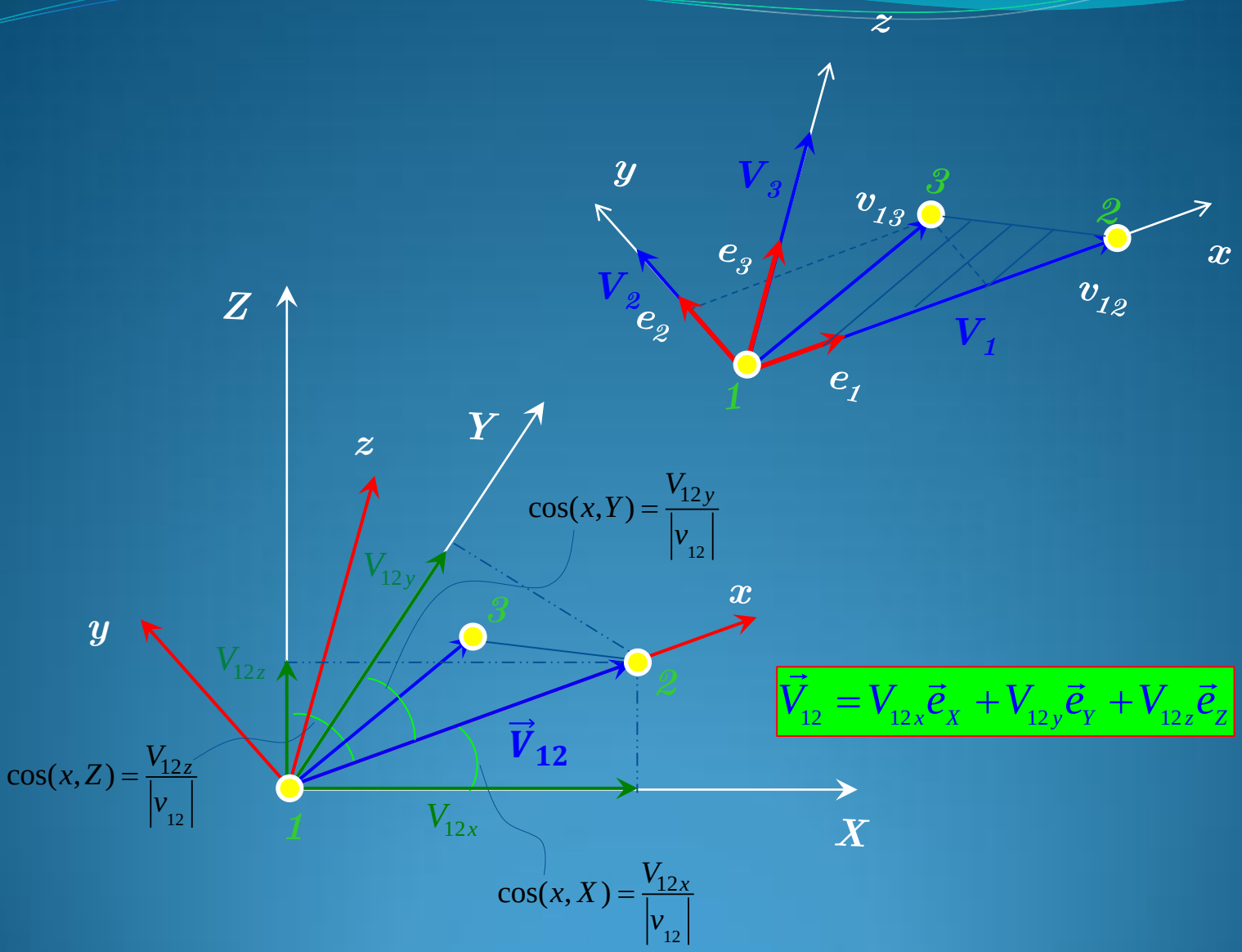
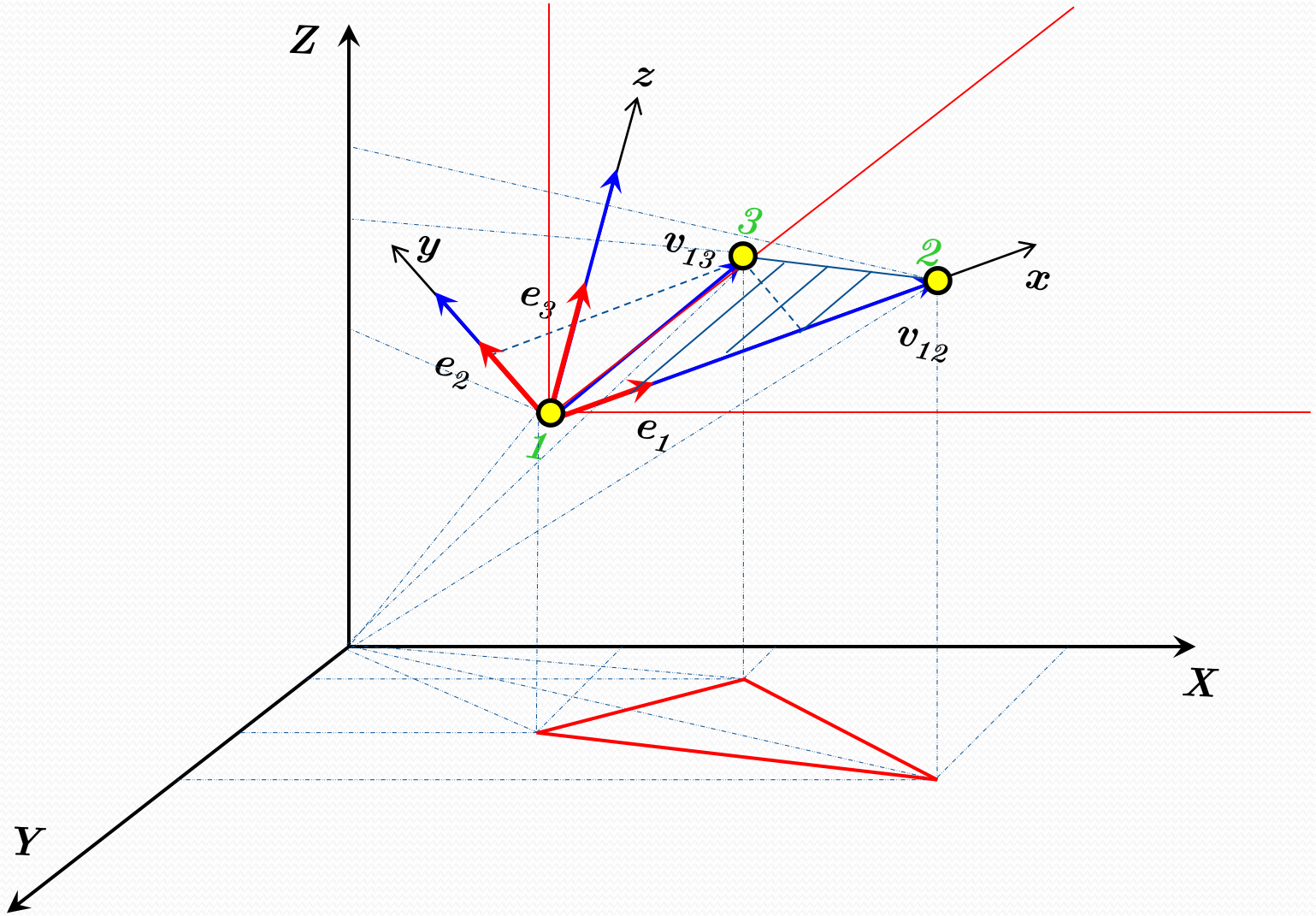
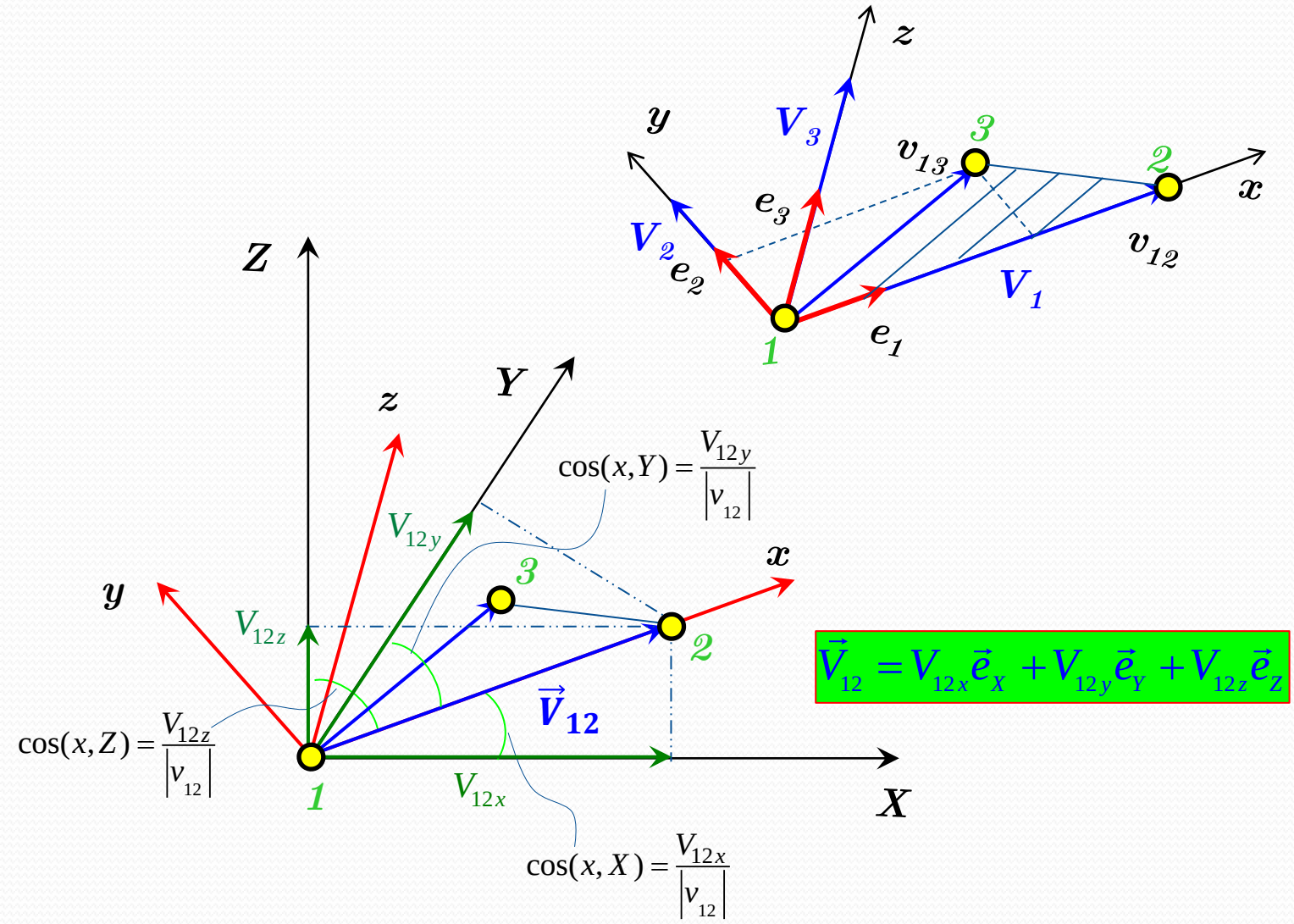

Z
z
y
x
3
2
1
$v_{13}$
$v_{12}$
$e_3$
$e_2$
$e_1$
X
Y


z
y
V3
v13
3
2
x
e3
V2
e2
v12
V1
1
e1
Z
Y
z
cos(x,Y) = V12y / |v12|
V12y
x
y
3
V12z
2
V12
cos(x,Z) = V12z / |v12|
1
V12x
X
cos(x,X) = V12x / |v12|
V12 = V12x eX + V12y eY + V12z eZ


Z
z
y
x
3
2
1
$v_{13}$
$v_{12}$
$e_3$
$e_2$
$e_1$
X
Y


z
y
V3
3
v13
2
x
e3
V2
e2
v12
V1
1
e1
Z
Y
z
cos(x,Y) = V12y / |v12|
V12y
x
3
y
V12z
2
V12
cos(x,Z) = V12z / |v12|
1
V12x
X
cos(x,X) = V12x / |v12|
V12 = V12x eX + V12y eY + V12z eZ